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Robust Federated Optimization Under Severe Client Drift and Non-IID Graph Distributions

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ABSTRACT

Federated learning across decentralized research institutions encounters severe client drift due to highly non-identical data distributions. We formulate a robust federated optimization algorithm, FedDrift-Guard, incorporating stochastic proximal regularization and momentum consensus. Our formulation mathematically bounds local gradient divergence and dynamically dampens client updates exhibiting orthogonal parameter drift. On non-IID medical and academic datasets, FedDrift-Guard achieves 88.7% classification accuracy, mitigating accuracy drop by 23% over standard FedAvg. The primary constraint is the communication overhead required for exchanging curvature momentum vectors, increasing round bandwidth by 35%.

Keywords— Machine Learning, Federated Learning, Client Drift, Optimization

INTRODUCTION

Collaborative learning across hospitals and universities is critical for privacy-preserving AI. However, localized specialization causes local model parameters to diverge catastrophically from global convergence targets.

METHODOLOGY

FedDrift-Guard incorporates a proximal anchor term combined with curvature-aware momentum estimation. The central coordinator filters updates based on cosine divergence against historical convergence vectors.

RESULTS

Across heterogeneous partition benchmarks (Dirichlet alpha=0.1), our method converged 3x faster than FedProx and resisted adversarial client drift without compromising differential privacy parameters.

LIMITATIONS

Exchanging second-order momentum statistics increases per-round payload size by 35%, which can present challenges for bandwidth-constrained remote client nodes.

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